Module 4

2 marks questions (Study material)

1. What is the main difference between supervised and unsupervised learning?
2. What is the objective of the K-means algorithm?
3. Name the key limitations of K-means?
4. Why is dimensionality reduction useful?
5. Give one real-world application of unsupervised learning.
6. Why is evaluating unsupervised models challenging?
7. What is a neural network?
8. What is an activation function? Give one example.
9. What is the role of the hidden layer in a neural network?
10. What is back propagation?
11. What is the purpose of a loss function?
12. What is over fitting in neural networks?
13. Name one type of specialized neural network and its use case.
14. What is a dropout layer?
15. Give one real-world application of neural networks.
16. What is the use of sigmoid function in ANN?
17. What is a perceptron?
18. What is the purpose of the bias term in a neural network?
19. What is the vanishing gradient problem?
20. What is the difference between batch and stochastic gradient descent?
21. What is a convolutional layer used for?
22. What is the purpose of pooling layers in CNNs?
23. What is transfer learning in neural networks?
24. What is a feed forward neural network?
25. What is the role of weights in a neural network?
26. Why are non-linear activation functions necessary?
27. What is the learning rate in gradient descent?
28. What is attention mechanism in neural networks?
29. Give an example of a neural network for image classification.
30. How are neural networks used in NLP?

5 marks questions (Book Bank)

RT ANSWER-TYPE Q

1. How unsupervised learning is different from supervised learning? Explain with some examples. (5)
2. Mention few application areas of unsupervised learning. (5)
3. What are the broad three categories of clustering techniques? Explain the characteristics of each briefly. (5)
4. Describe how the quality of clustering is measured in the k-means algorithm? (5)
5. Describe the main difference in the approach of k-means and k-medoids algorithms with a neat diagram.
6. What is a dendrogram? Explain its use. (5)
7. What is SSE? What is its use in the context of the k-means algorithm? (5)
8. Explain the k-means method with a step-by-step algorithm. (5)
9. Describe the concept of single link and complete link in the context of hierarchical clustering.
10. How apriori principle helps in reducing the calculation overhead for a market basket analysis? Provide an example to explain. (5)

10 marks questions (Book Bank)

LONG ... (10 MARKS EACH)

1. You are given a set of one-dimensional data points: {5, 10, 15, 20, 25, 30, 35}. Assume that $k = 2$ and first set of random centroid is selected as {15, 32} and then it is refined with {12, 30}.
   (a) Create two clusters with each set of centroid mentioned above following the k-means approach
   (b) Calculate the SSE for each set of centroid (10)
2. Explain how the Market Basket Analysis uses the concepts of association analysis. (10)
3. Explain the Apriori algorithm for association rule learning with an example.
4. How the distance between clusters is measured in hierarchical clustering? Explain the use of this measure in making decision on when to stop the iteration.
5. How to recompute the cluster centroids in the k-means algorithm? (10)
6. Discuss one technique to choose the appropriate number of clusters at the beginning of clustering exercise. (10)
7. Discuss the strengths and weaknesses of the k-means algorithm. (10)
8. Explain the concept of clustering with a neat diagram.
9. During a research work, you found 7 observations as described with the data points below. You want to create 3 clusters from these observations using K-means algorithm. After first iteration, the clusters C1, C2, C3 has following observations:

   C1: {(2,2), (4,4), (6,6)}

   C2: {(0,4), (4,0)}

   C3: {(5,5), (9,9)}

   If you want to run a second iteration then what will be the cluster centroids? What will be the SSE of this clustering? (10)
10. In a software project, the team is trying to identify the similarity of software defects identified during testing. They wanted to create 5 clusters of similar defects based on the text analytics of the defect descriptions. Once the 5 clusters of defects are identified, any new defect created is to be classified as one of the types identified through clustering. Explain this approach through a neat diagram. Assume 20 Defect data points which are clustered among 5 clusters and k-means algorithm was used.